

SmolVLM

Redefining Small and Efficient Multimodal Models

Hugging Face · Stanford University · arXiv 2504.05299

256M – 2.2B

Parameters

< 1 GB RAM

Smallest Model

300×

Smaller than Llama-3.1-80B

Fully Open

Weights · Data · Code

Large VLMs Are Powerful but Impractical for Edge Deployment

The Problem

- State-of-the-art VLMs (GPT-4V, LLaVA-1.5, Idefics-80B) require 10–80+ GB GPU RAM, making mobile/edge deployment infeasible
- On-device inference demands sub-1 GB memory footprints with real-time throughput (>50 tokens/sec)
- Existing efficient models sacrifice too much accuracy or remain too large for true edge deployment

Idefics-80B

80B params

RAM: ~160 GB

Molmo-A1B-7B

~7B params

RAM: 27.7 GB

SmolVLM-256M

256M params

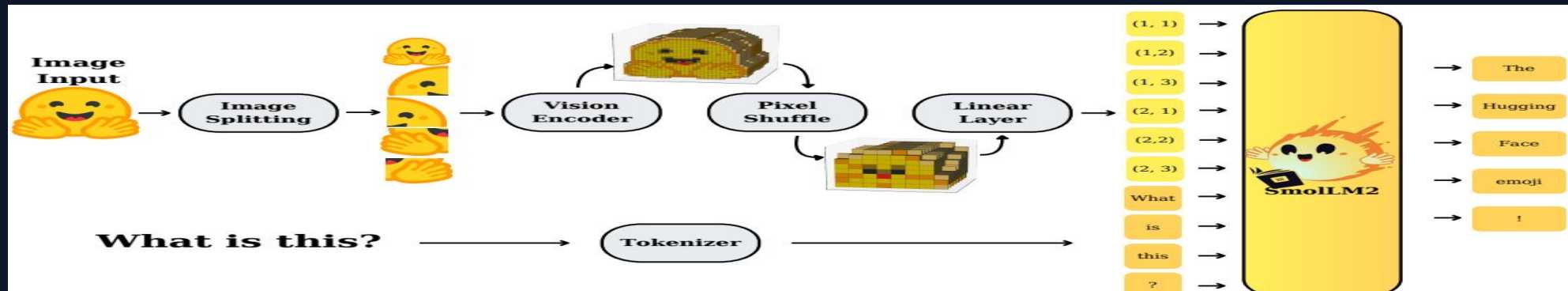
RAM: 0.8 GB

SmolVLM-256M surpasses the 300× larger Idefics-80B — proving architectural efficiency matters more than raw scale.

SmolVLM Architecture Pipeline

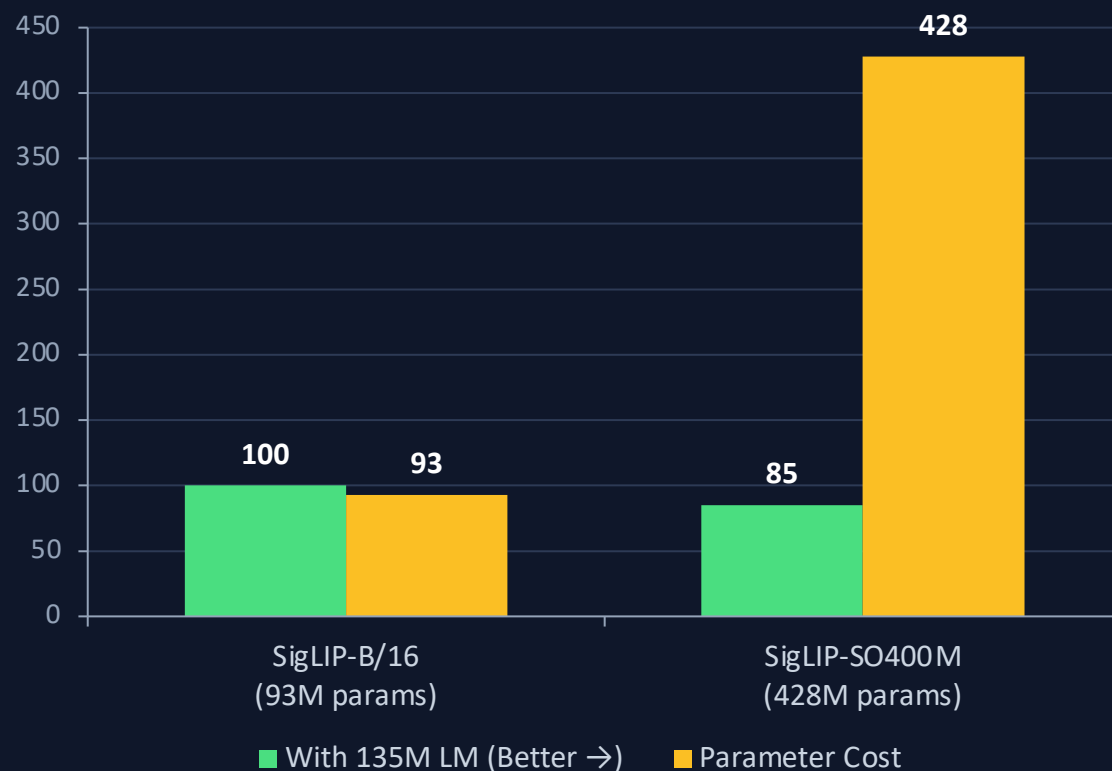


Video support: Frames are sampled and processed through the same pipeline. Text tokens are concatenated with visual embeddings.



Smaller Vision Encoders Outperform Larger Ones in Compact VLMs

Finding 1: A 93M SigLIP-B/16 paired with a 135M LM outperforms the 428M SigLIP-SO400M with the same LM. The larger encoder only justifies its 10% parameter overhead at $\geq 1.7\text{B}$ LM scale.



93M Encoder

SigLIP-B/16 achieves higher accuracy with the 135M LM despite being 4.6× smaller than SigLIP-SO400M

428M Encoder

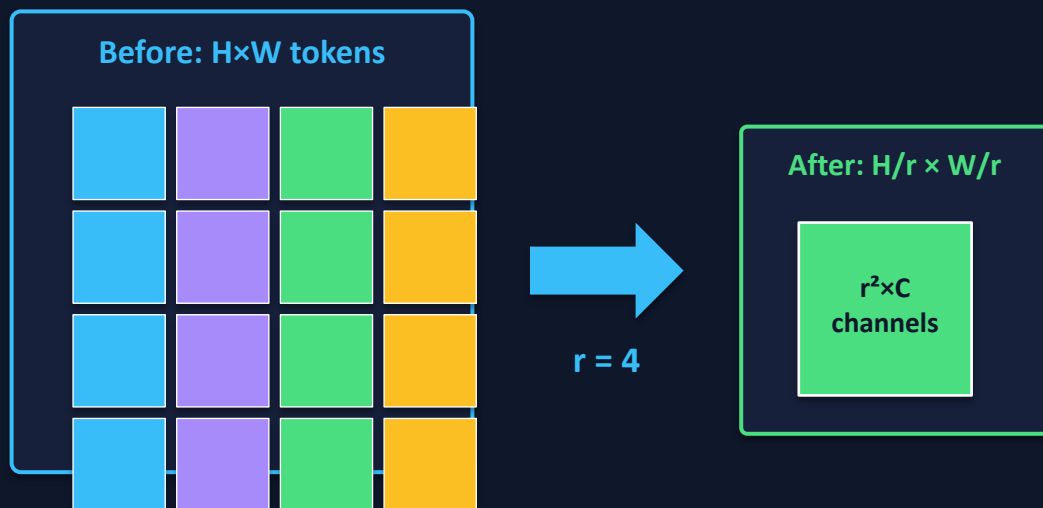
SigLIP-SO400M only justifies its extra parameters when paired with $\geq 1.7\text{B}$ language models

Design Rule

Match encoder capacity to LM size — over-investing in vision parameters hurts small VLMs

Aggressive Pixel Shuffle ($r=4$) Cuts Visual Tokens 16×

Pixel Shuffle Mechanism (Space-to-Depth)



Spatial features rearranged into channel dimension $\rightarrow$ 16× fewer tokens at $r=4$

Compression Factor Comparison

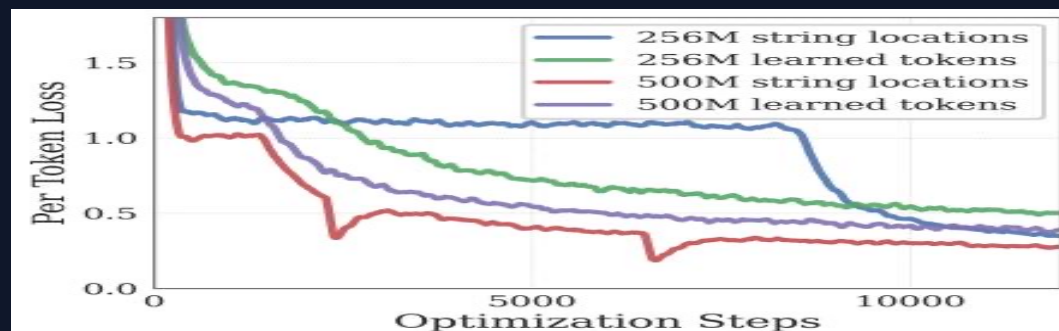
$r = 2$ **4× reduction**

Preferred for large VLMs ($\geq 1.7B$)

$r = 4$ **16× reduction**

Optimal for small VLMs (135M–500M)

$r=4$ eases attention overhead and improves long-context modeling for small VLMs by drastically reducing sequence length.



Extending Context from 2K to 16K Tokens Unlocks Higher Resolution



RoPE Base Adjustment

Increased from 10K to 273K to support longer positional encodings without degradation

Mixed-Length Training

Fine-tuned on mixed long/short-context data to maintain performance across sequence lengths

Resolution Impact

Longer context enables processing more sub-images at higher resolution per image

Benchmark Performance: Competitive at Sub-1GB to 4.9GB RAM

Category	Benchmark	SmolVLM 256M	SmolVLM 500M	SmolVLM 2.2B	Best Efficient OS
Single-Image	OCRBench	52.6%	61.0%	72.9%	54.7% (Molmo)
Single-Image	AI2D	46.4%	59.2%	70.0%	71.0% (Molmo)
Single-Image	ChartQA	55.6%	62.8%	68.7%	48.0% (Molmo)
Single-Image	TextVQA	50.2%	60.2%	73.0%	61.5% (Molmo)
Single-Image	DocVQA	58.3%	70.5%	80.0%	77.7% (Molmo)
Single-Image	ScienceQA	73.8%	80.0%	89.6%	87.5% (Molmo)
Multi-task	MMMU	29.0%	33.7%	42.0%	33.9% (Molmo)
Multi-task	MathVista	35.9%	40.1%	51.5%	37.6% (Molmo)
Multi-task	MMStar	34.6%	38.3%	46.0%	43.1% (Molmo)
Video	Video-MME	33.7%	42.2%	52.1%	45.0% (InternVL2)
Video	MLVU	40.6%	47.3%	55.2%	48.2% (InternVL2)
SmolVLM-256M					
SmolVLM-500M					
SmolVLM-2.2B					
Molmo-A1B-7B					
RAM: 0.8 GB		RAM: 1.2 GB		RAM: 4.9 GB	
				RAM: 27.7 GB	

Real-Time On-Device Inference on Apple Silicon

HuggingSnap: Fully Local iOS Inference

~80 tok/s

SmolVLM-256M on M1 Max 64GB

Pro desktop throughput

~55 tok/s

SmolVLM-256M on MacBook Air M3 16GB

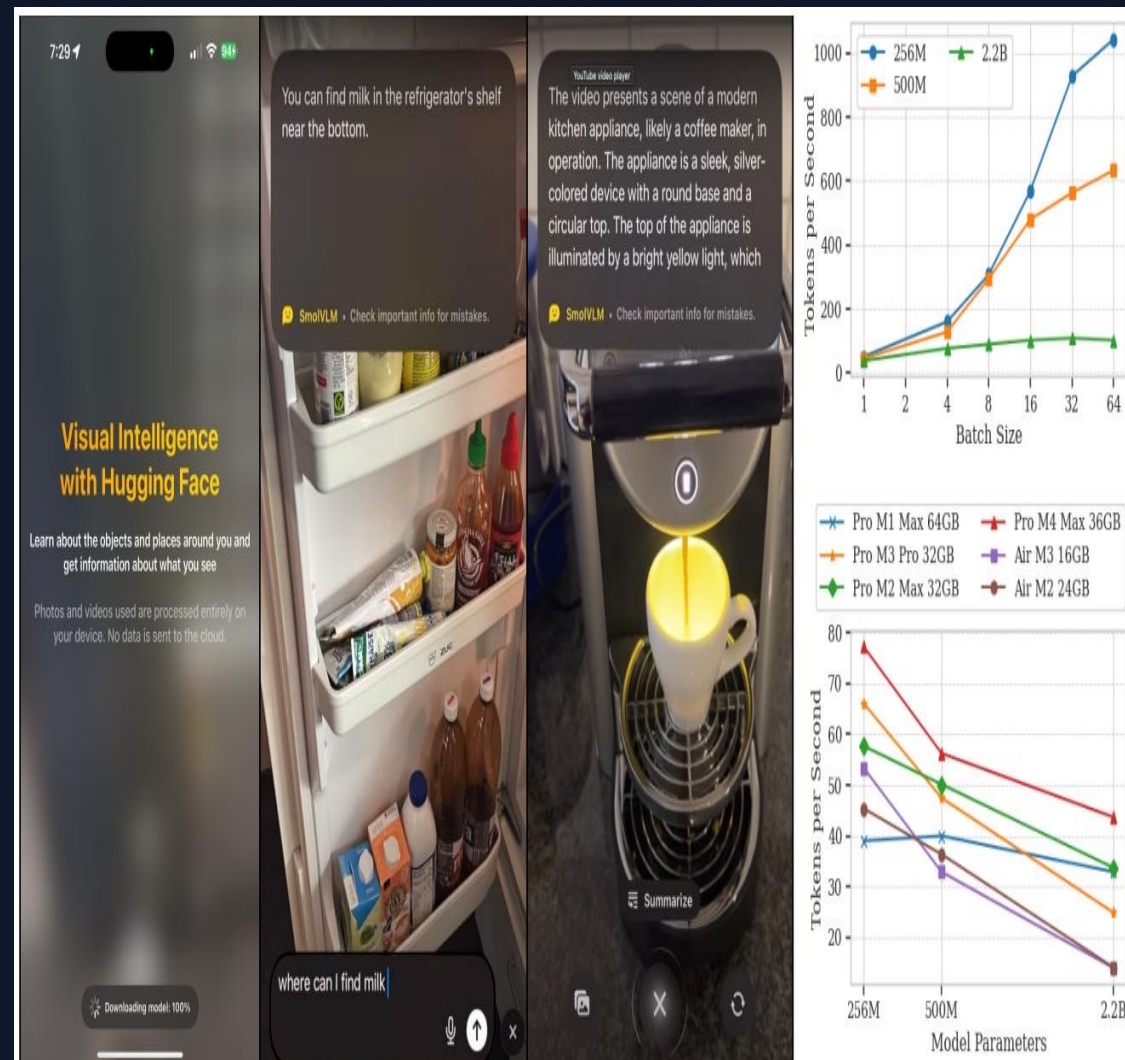
Consumer laptop throughput

~1000 tok/s

SmolVLM-256M at batch size 64

Peak throughput on high-end devices

**Throughput scales linearly with batch size for the 256M model.
No cloud dependency — fully on-device processing.**



Key Takeaways: Design Principles for Efficient Multimodal Models

1

Balance Encoder–LM Compute

Match vision encoder capacity to language model size. A 93M encoder outperforms a 428M encoder for small LMs — over-investing in vision parameters hurts compact VLMs.

2

Extend Context Length

Increase RoPE base from 10K to 273K and fine-tune on mixed long/short data. The jump from 2K to 16K context significantly improves high-resolution image understanding.

3

Use Aggressive Token Compression

Pixel shuffle at $r=4$ delivers 16× token reduction. Small models benefit disproportionately from aggressive compression by reducing attention overhead.

4

Sub-1GB Inference Is Achievable

SmolVLM-256M runs at 0.8 GB GPU RAM and surpasses the 300× larger LLaVA-72B — architectural efficiency matters more than raw parameter count.

5

Open-Source Everything

All weights, datasets, code, and a mobile app (HuggingSnap) are fully released to promote reproducibility and accelerate on-device AI research.